



Introduction to Bioengineering Theory

(BBL1020; 3-0-0)

Lecture 4

Chemical basis of life

Biomolecules

Evolution and Taxonomy

**.....is the
basis of life**

**Chemistry is the
basis of life**

Chemical Basis of life

“The origin of life presents a particularly difficult problem for chemistry because it is essentially unconstrained: there are a very, very large number of reactions that might have occurred on prebiotic Earth, and few hard limits on the types of compounds that might have been available, on the possible reaction conditions, and on the processes that might finally have led to elementary cells of the type that we now know.”

From Whitesides Research Group, Harvard University

“What is, however, clear is the central conundrum of understanding “life”: that is, that a living cell is composed of molecules; that molecules in cells are reacting; that neither the molecules nor the reactions are individually alive; but that the cell – as a collection of reactions and processes - is alive.

Several theories of the origins of life propose that life emerged spontaneously from the self-assembly of organic reactions, (im)probably occurring chaotically in complex mixtures of molecules. The chemistry that permit the assembly of simple organic reactions into networks with complex emergent behaviors, however, remain incompletely understood.

Conditions were different from today

- Different temperatures, types of molecules, number of molecules, pressure, charge density: possible of reactions was vast – small molecules – complex molecules – structure entity – organism



Prof. George M. Whitesides
Department of Chemistry
and
Chemical Biology
Harvard University

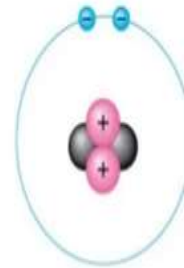
Chemical Basis of life

Chemicals play many roles in an organism life:

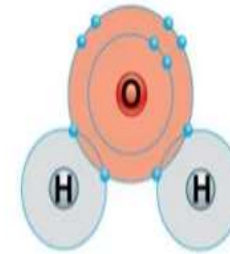
1. Organisms use chemical signals for **muscle movements, mating and defense.**
2. DNA is a combination of chemical elements
3. Chemical reactions of body occur in cells consisting of **70-95% water.**

Organism are composed of elements

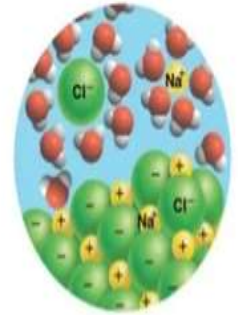
Chapter 2: Big Ideas



Elements, Atoms, and Compounds



Chemical Bonds



Water's Life-Supporting Properties



Element	% by mole
Hydrogen	63.0
Oxygen	26.0
Carbon	9.0
Nitrogen	1.25
Calcium	0.25
Phosphorus	0.19
Potassium	0.06
Sulfur	0.06
Sodium	0.04
Chlorine	0.025
Magnesium	0.013
Iron	0.00004
Iodine	0.000002

99% of atoms in a human body come from these 4 elements

Origin and evolution of life

How?

Panspermia Theory

This theory proposed that life developed elsewhere in the universe and then arrived on earth. Microbial spores from space were supposedly brought to earth by asteroids or meteors; which jump-started life on earth. However, this theory states how life spread, not how life began. The validity of this theory is fiercely debated since no conclusive proof exists. Furthermore, this theory ignores the fact that space is incredibly vast, so travelling between planets would take thousands and thousands of years. It would be highly improbable for a lifeform to remain dormant or alive for such long durations.

When? Originated about 3.7 billion years ago



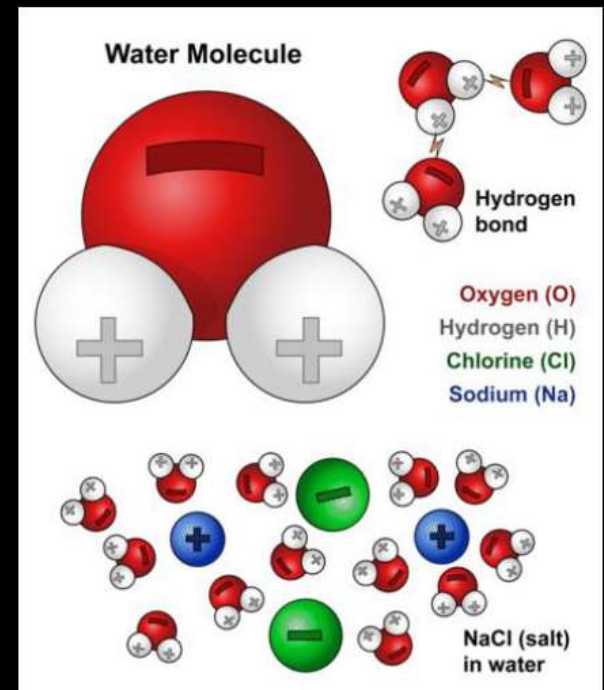
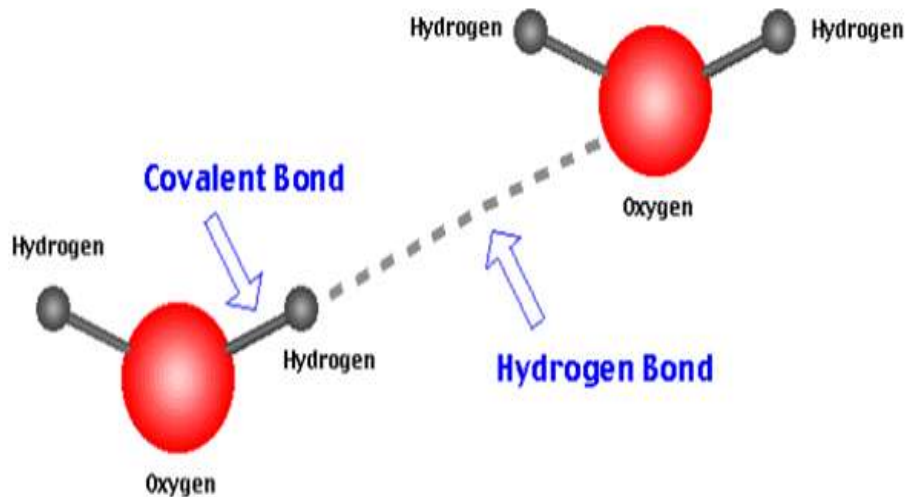
What is Panspermia Theory?

<https://www.youtube.com/watch?v=X1MBuUvddf8>

Water sustains life



A simple small molecule yet governs all form of the life

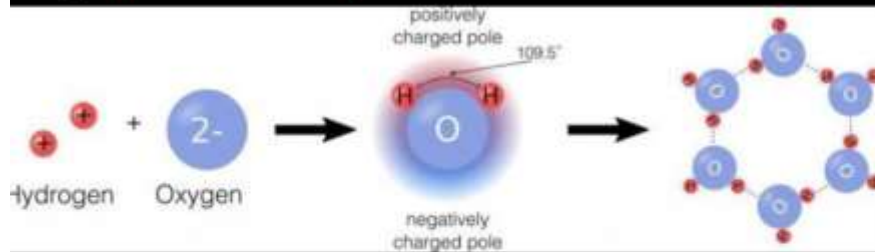


Molecule which can be present naturally in three distinct phases : Solid, Liquid and Gas.

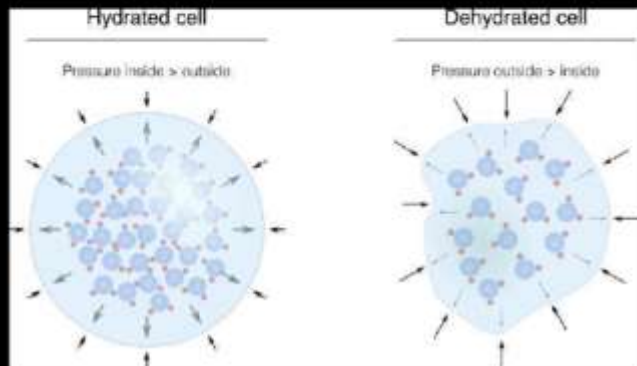
Water likes polar molecules and hates non-polars (hydrophobics)

Biological Role of Water

1. Molecular make-up of water: asymmetric and polar

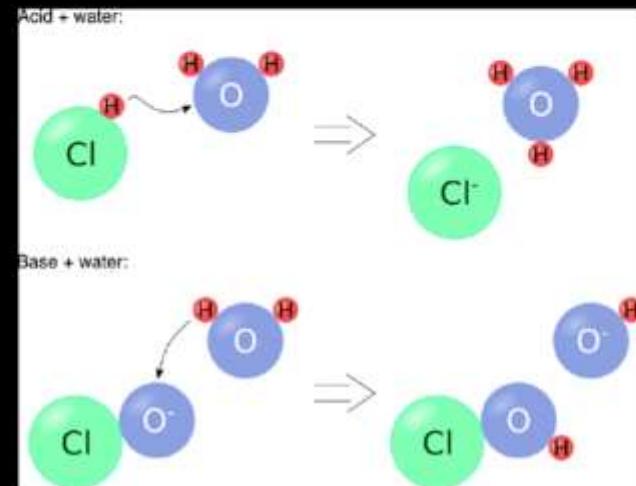


2. Supports cellular structure



3. Universal solvent: cohesion, molecules and ions interactions

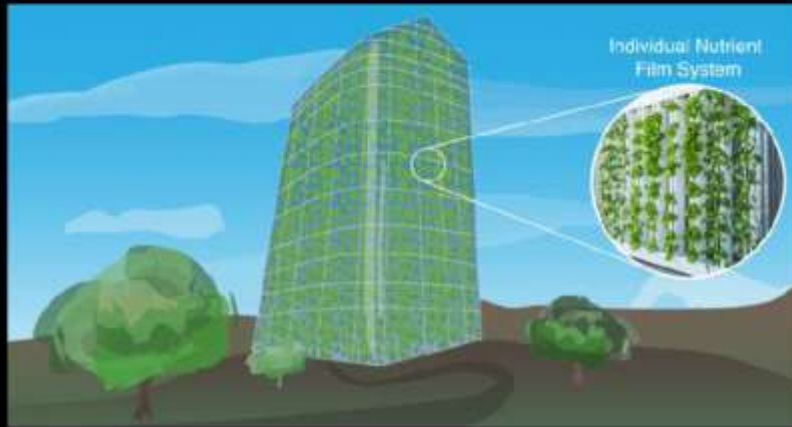
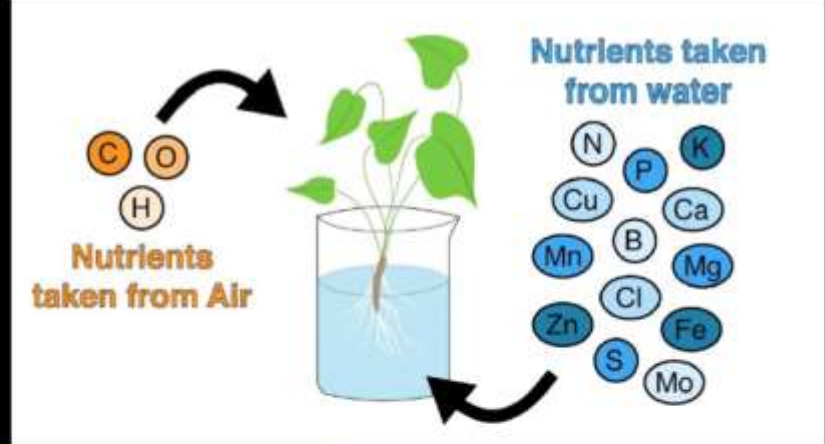
4. Facilitates chemical reactions: buffer



Hydroponics: The power of water to grow food



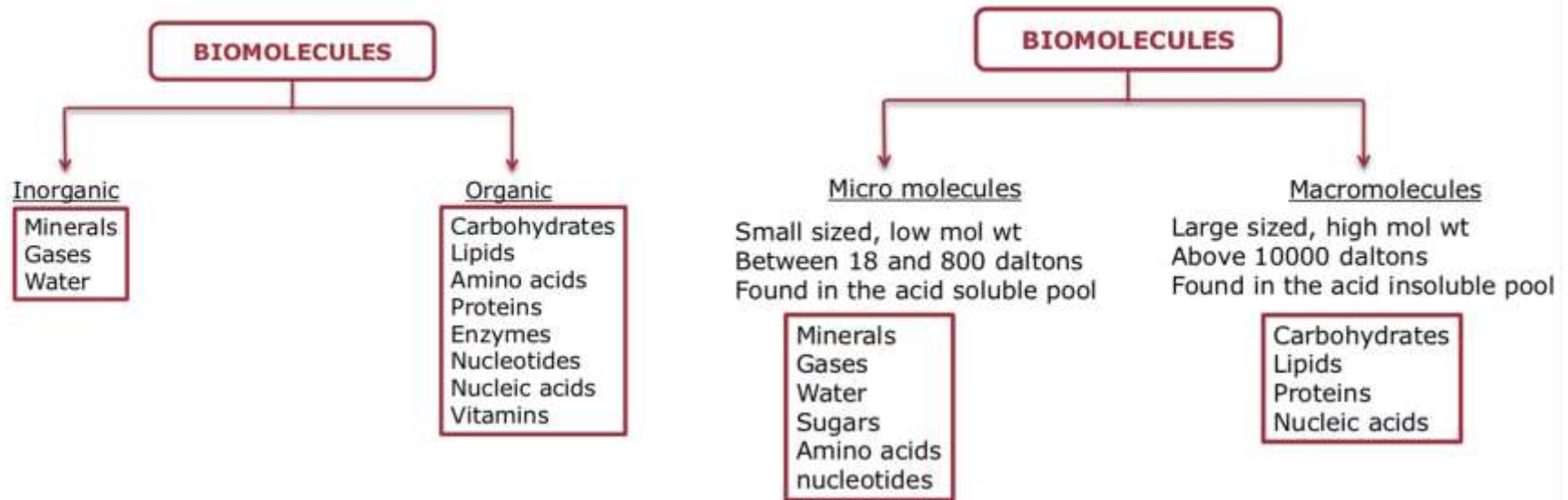
pulley-system of
water from the river



Vertical Farming

Biomolecules

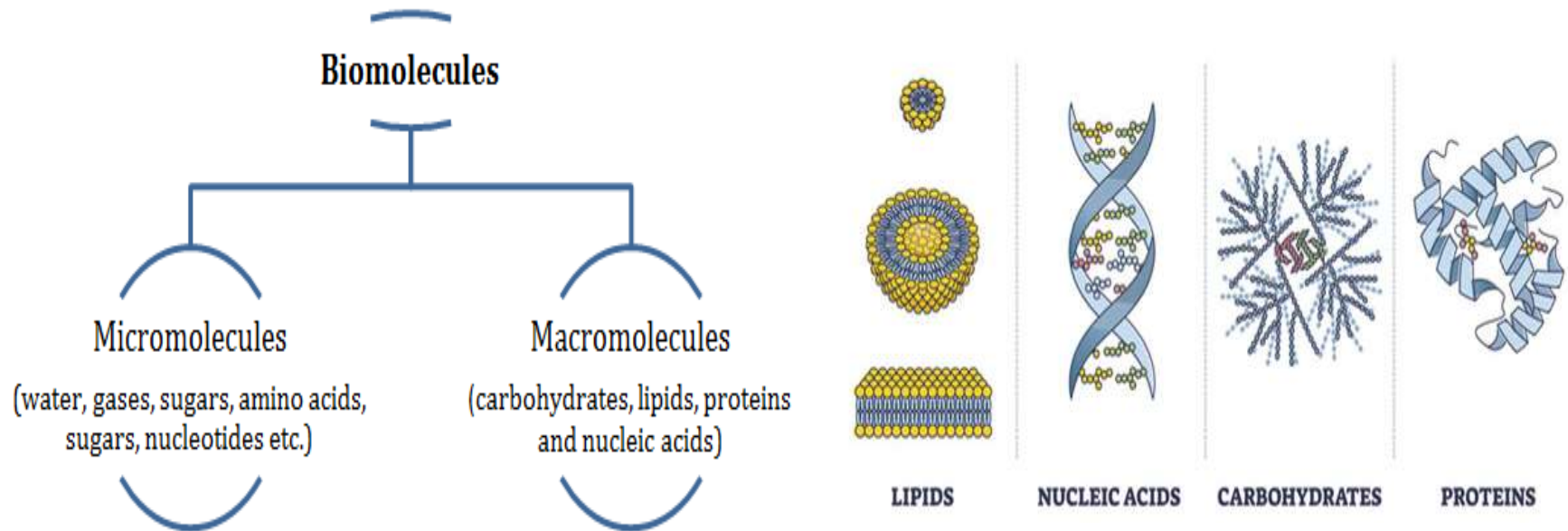
Chemical or Molecules present in the living organisms are called as Biomolecules. They include organic and inorganic compounds.



A molecule having a relatively low molecular weight is classified as Micromolecule, whereas large molecules (>1000Da) usually made up of polymerization of smaller molecules are classified as Macromolecules.

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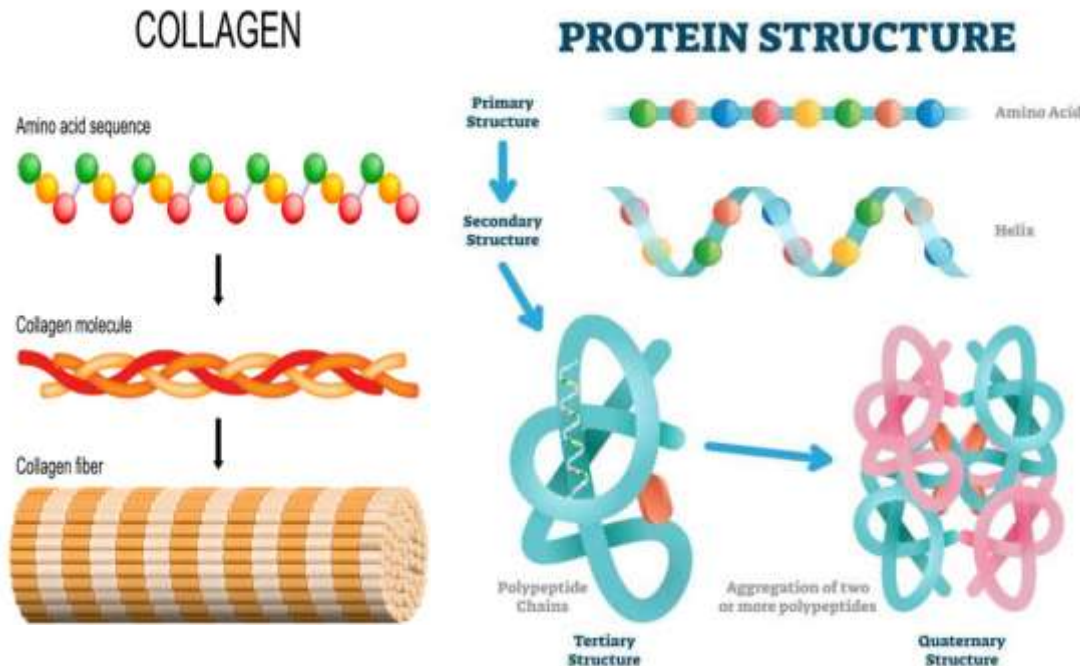
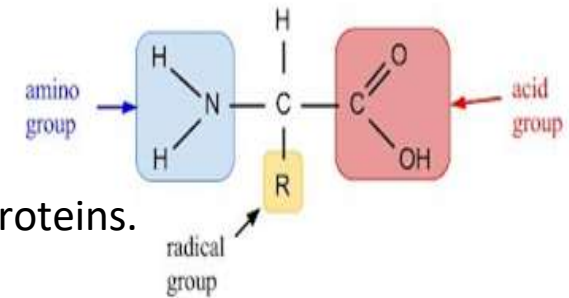


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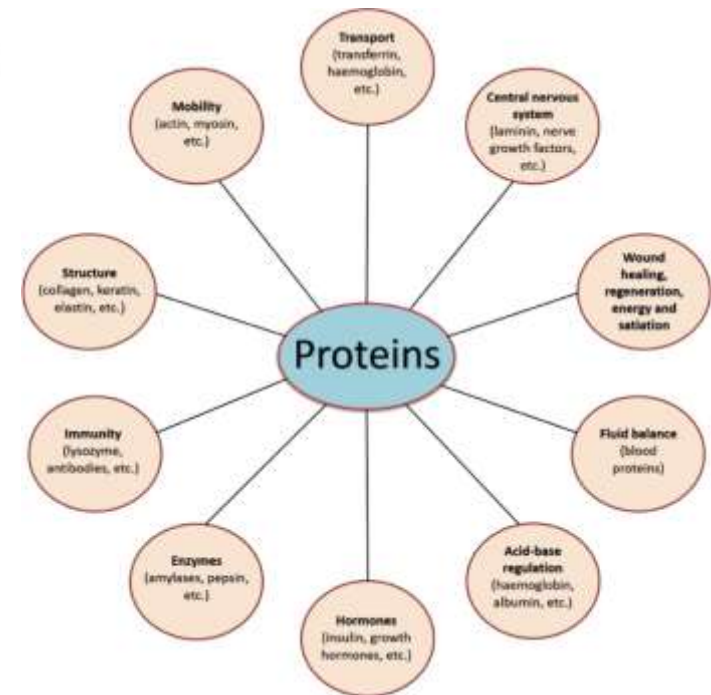
Biological macromolecules

1. Proteins Proteins are the heteropolymers of amino acids, linked together by peptide bond.

There are 20 standard amino acids, found in the structure of proteins.



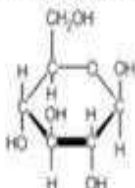
Most abundant animal protein

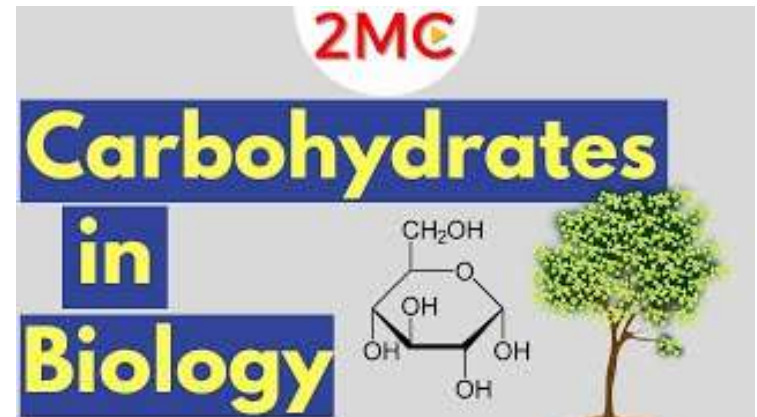


Functions of Protein

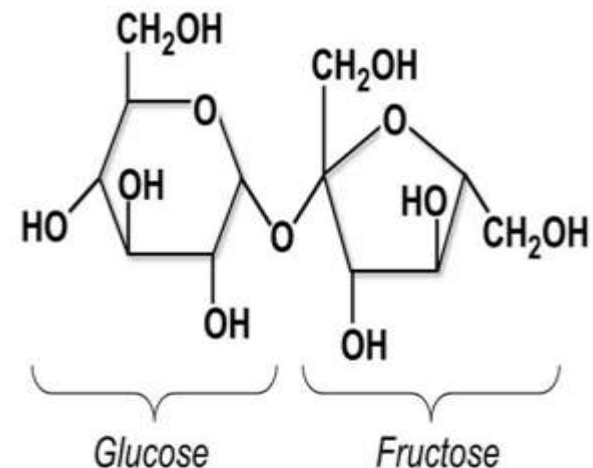
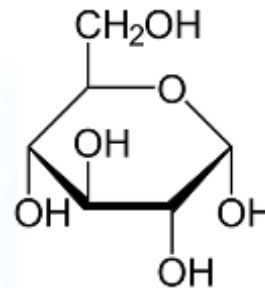
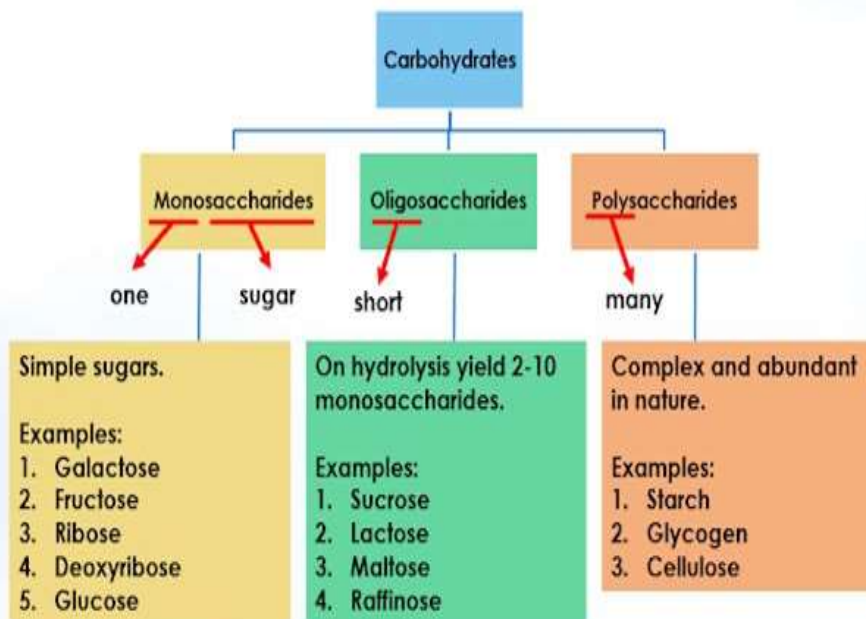
Biological macromolecules

2. Carbohydrates These are the most abundant organic molecules in nature and are the primary source of energy.

Biomolecules	Monomer(s)	Function	Food Source	Example(s)
Carbohydrates 	Monosaccharides	Provide material to build cell membrane Quick energy for cells	Pastas Breads Fruits Vegetables	Glucose Fructose Lactose Cellulose



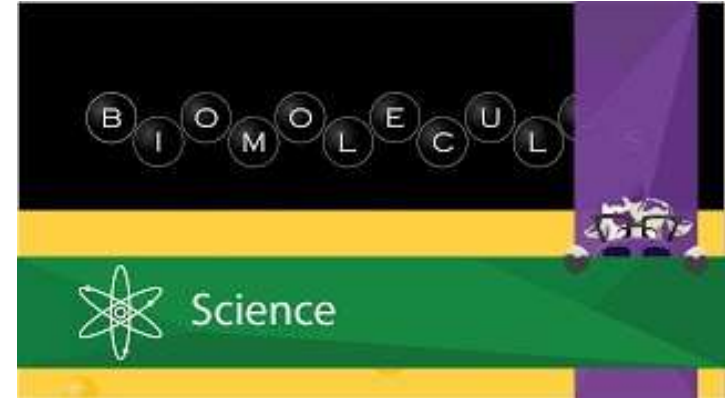
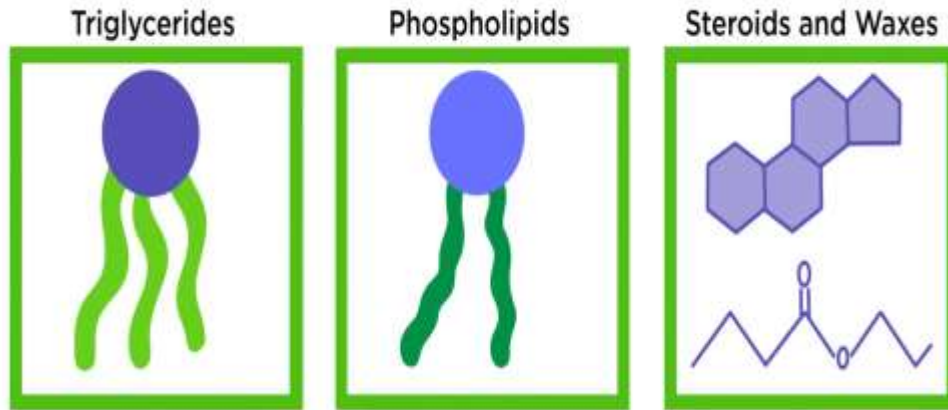
[watch here](#)



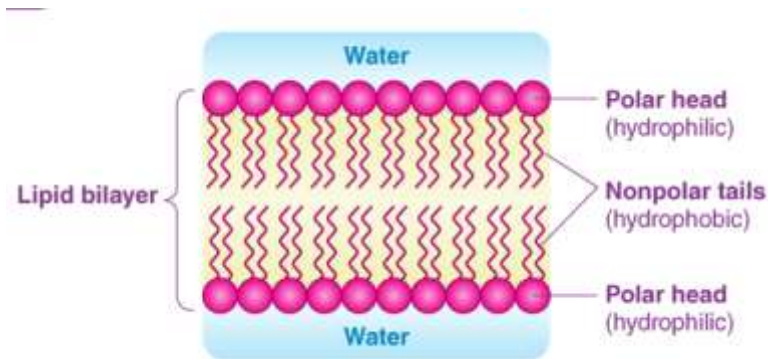
Biological macromolecules

3. Lipids Lipids are made up of fatty acid monomers which is responsible for energy storage, cell structural integrity and cellular signalling.

Main types of lipids:

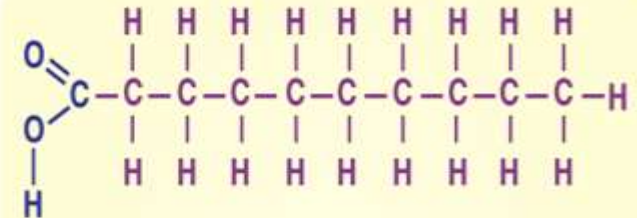


[Watch here](#)

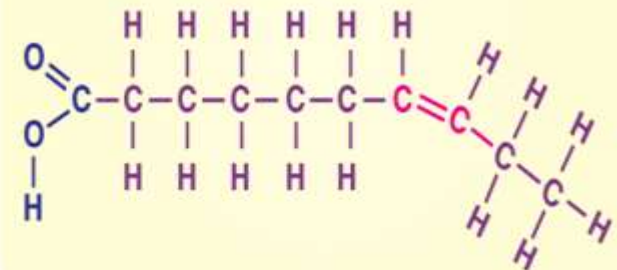


Phospholipids are responsible for cell membrane structure

(a) Saturated

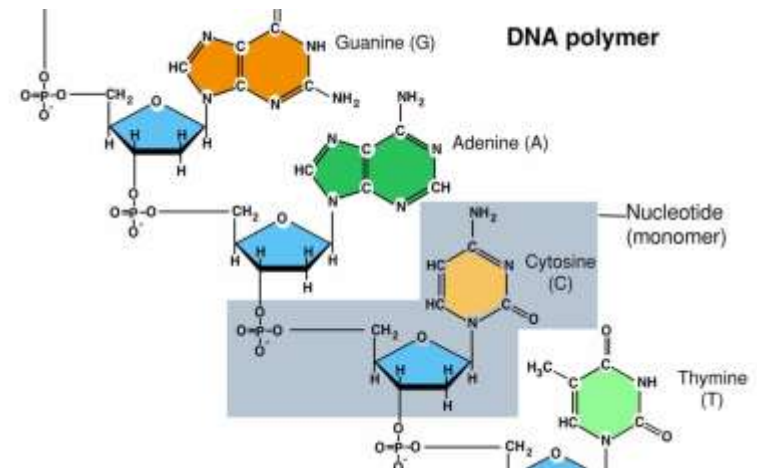
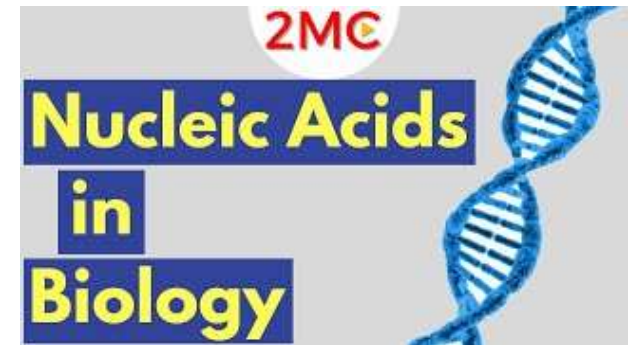
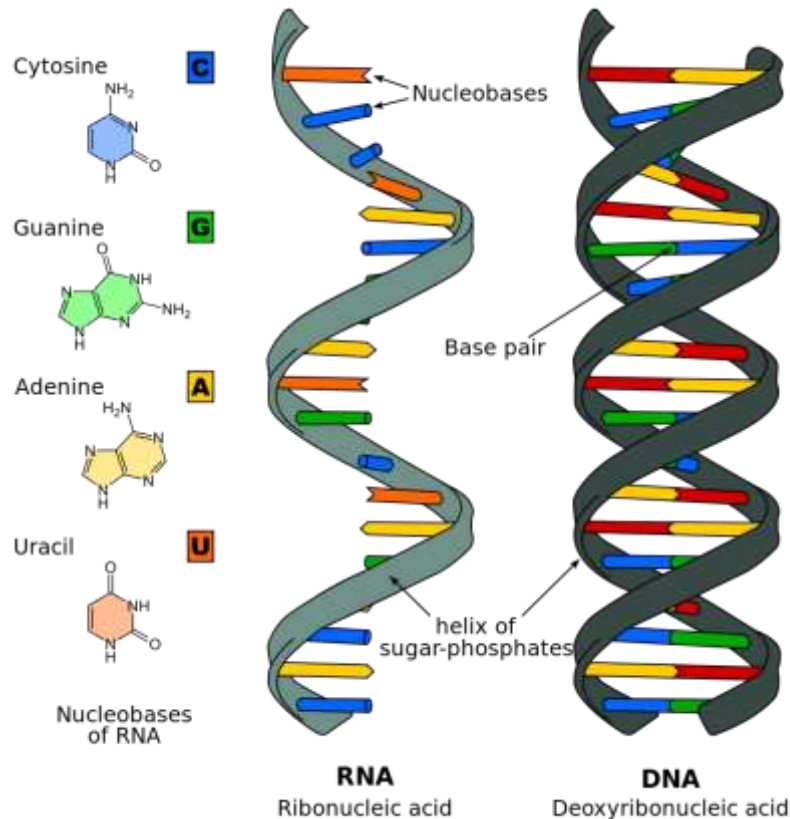


(b) Unsaturated



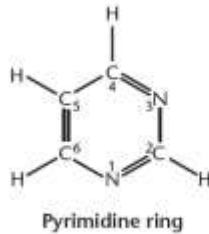
Biological macromolecules

4. **Nucleic Acids** Nucleic acids are made up of pentose sugar, nitrogenous base and phosphate group.

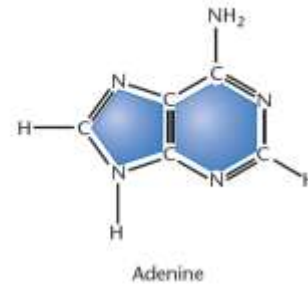
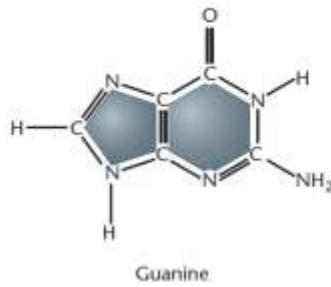
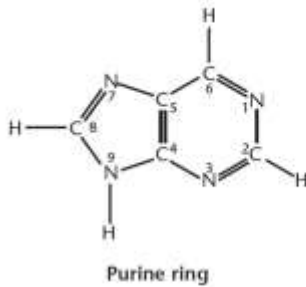


Building blocks of DNA & RNA

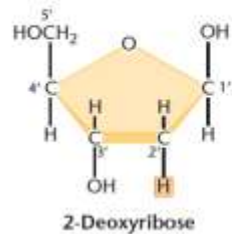
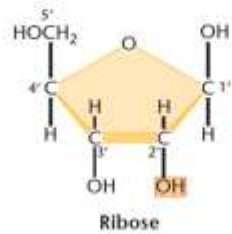
(a)



(b)



(b)



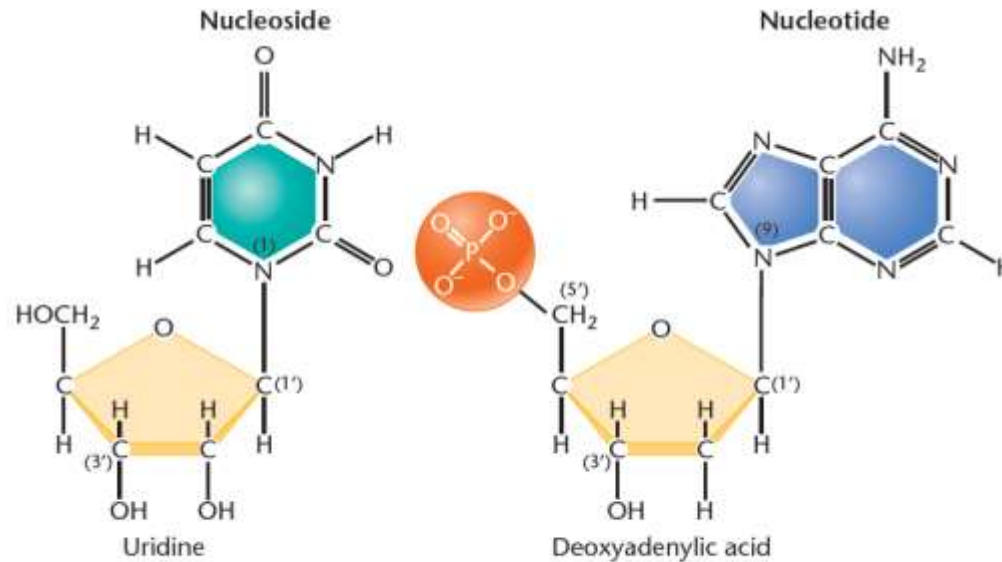
A 6 amino

G 2 amino 6 keto

C 2 keto 4 amino

U 2 keto 4 keto

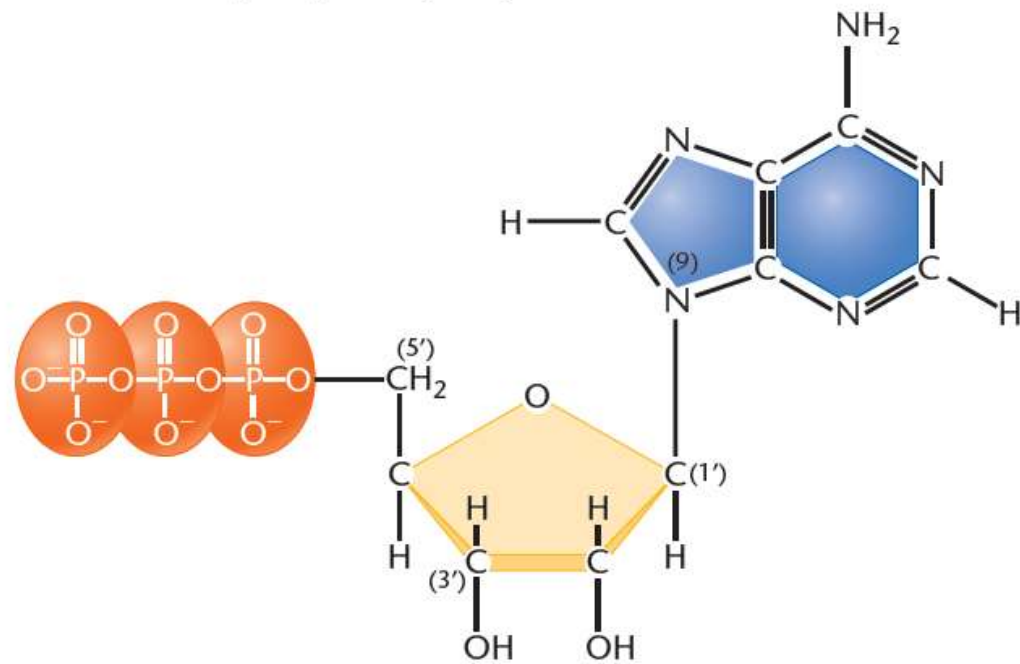
T 2 keto 4 keto 5 methyl



Know correct names of bases, nucleosides and nucleotides

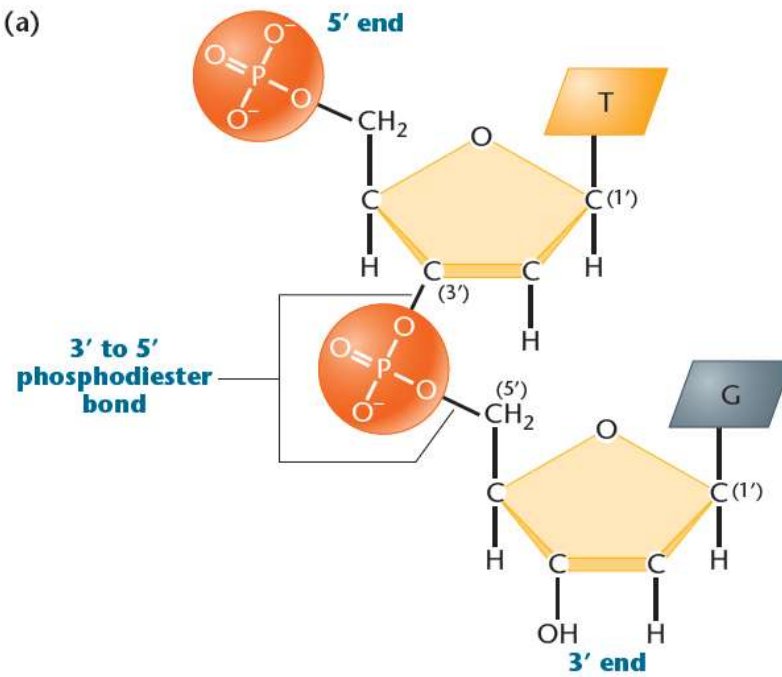
Ribonucleosides	Ribonucleotides
Adenosine Cytidine Guanosine Uridine	Adenylic acid Cytidylic acid Guanylic acid Uridylic acid
Deoxyribonucleosides	Deoxyribonucleotides
Deoxyadenosine Deoxycytidine Deoxyguanosine Deoxythymidine	Deoxyadenylic acid Deoxycytidylic acid Deoxyguanylic acid Deoxythymidylic acid

Nucleoside triphosphate (NTP)



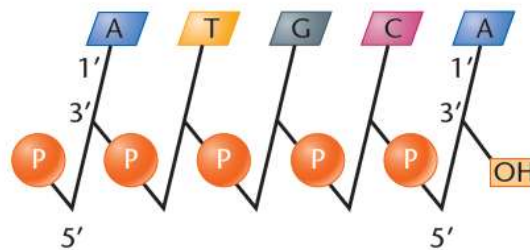
Adenosine triphosphate (ATP)

(a)



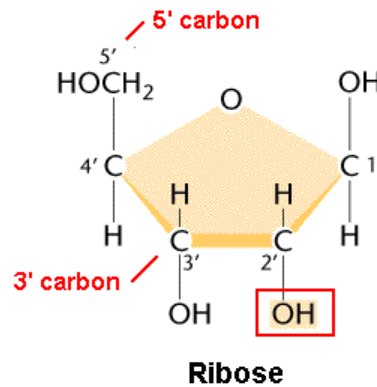
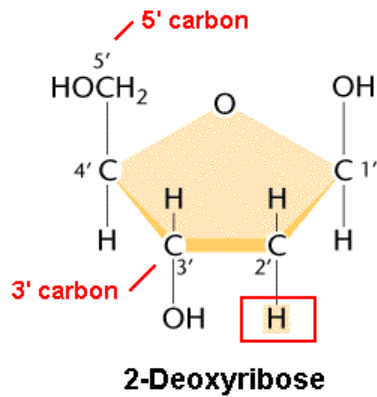
Phosphodiester bond

(b)



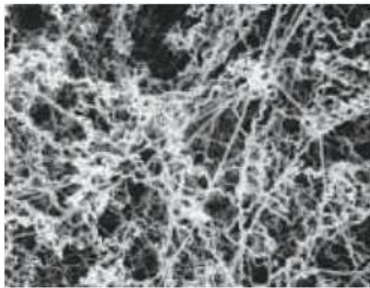
Why DNA is more stable than RNA?

Hydrogen atom at 2' position in deoxyribose of DNA makes it more stable than OH at 2' position in ribose of RNA



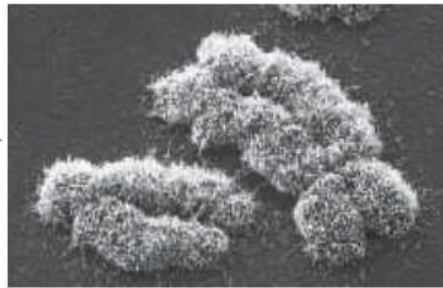
(Klug & Cummings 1997)

Electron micrographs of eukaryotic DNA in different stages of the cell cycle



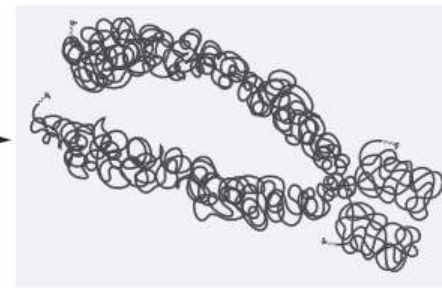
Interphase

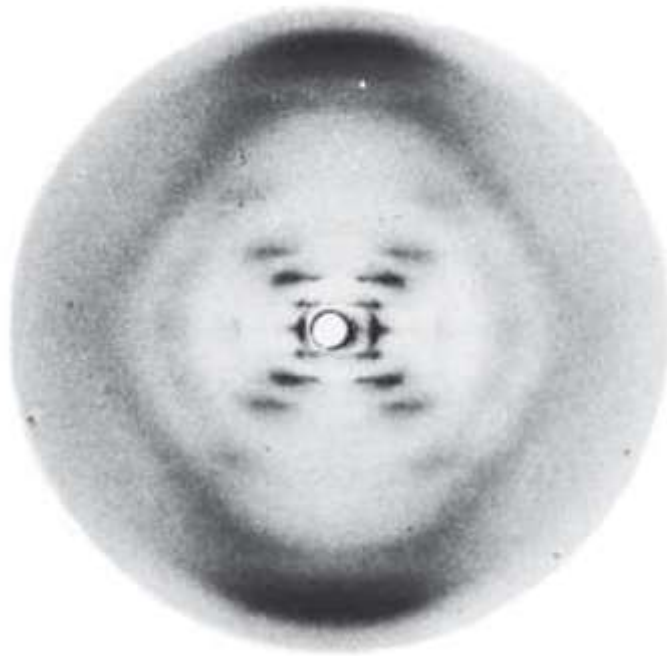
DNA is seen
as strands



Metaphase

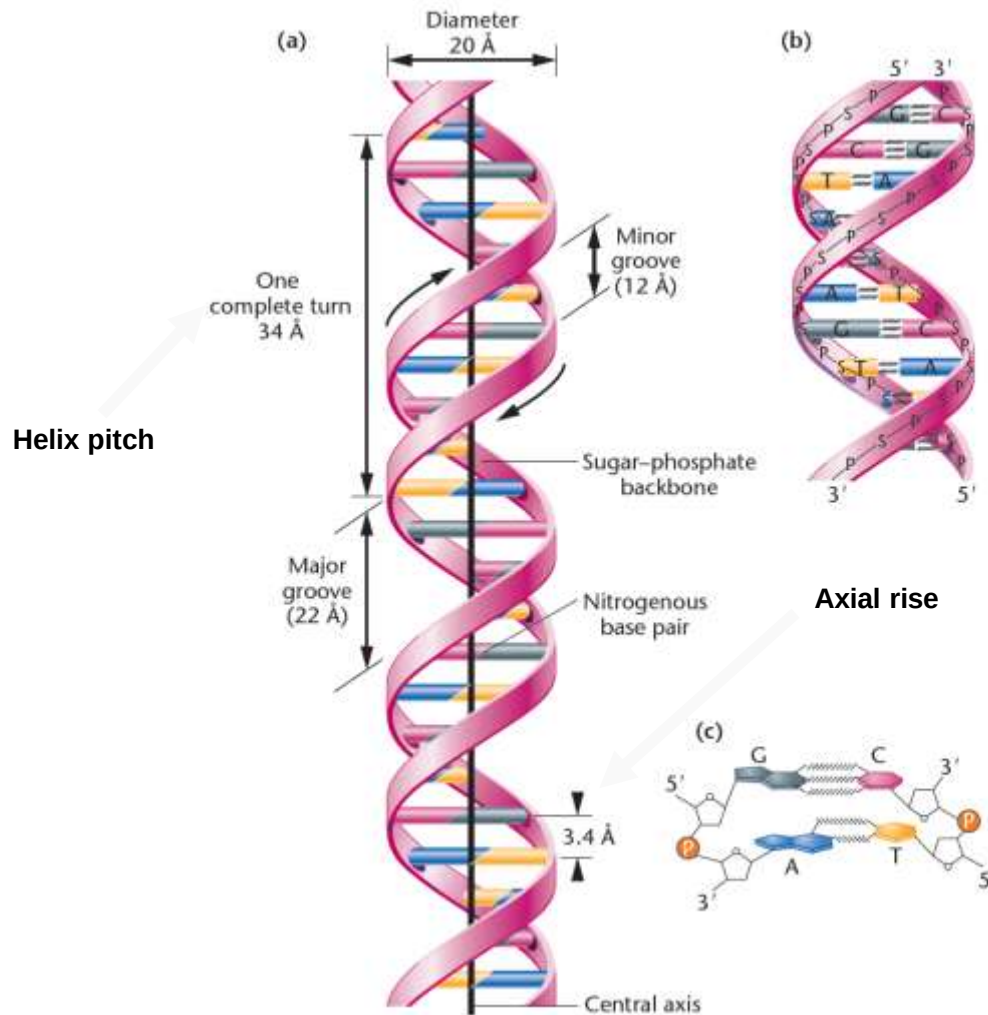
DNA is coiled as
chromosomes





X-ray crystal
structure of DNA by
Rosalind Franklin

Watson and Crick's model

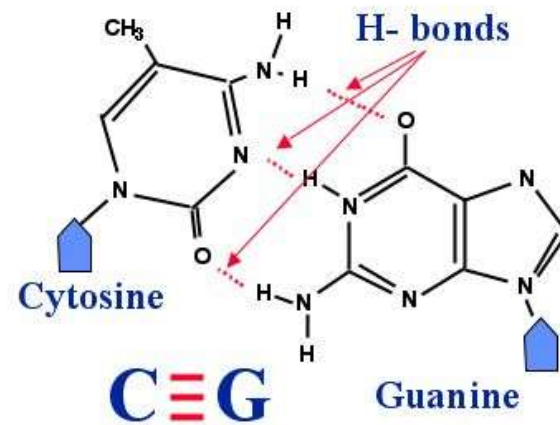
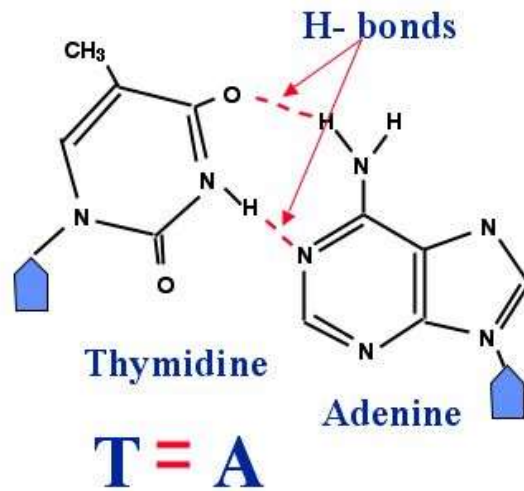


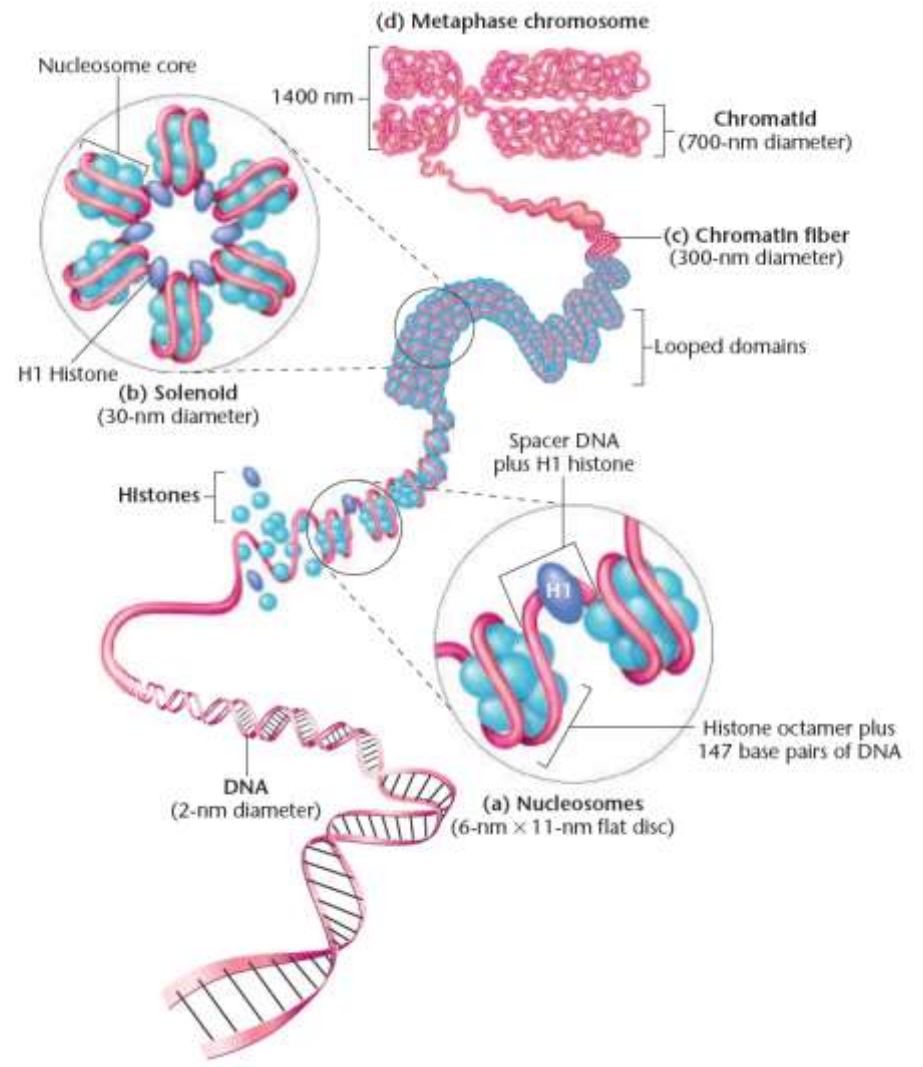
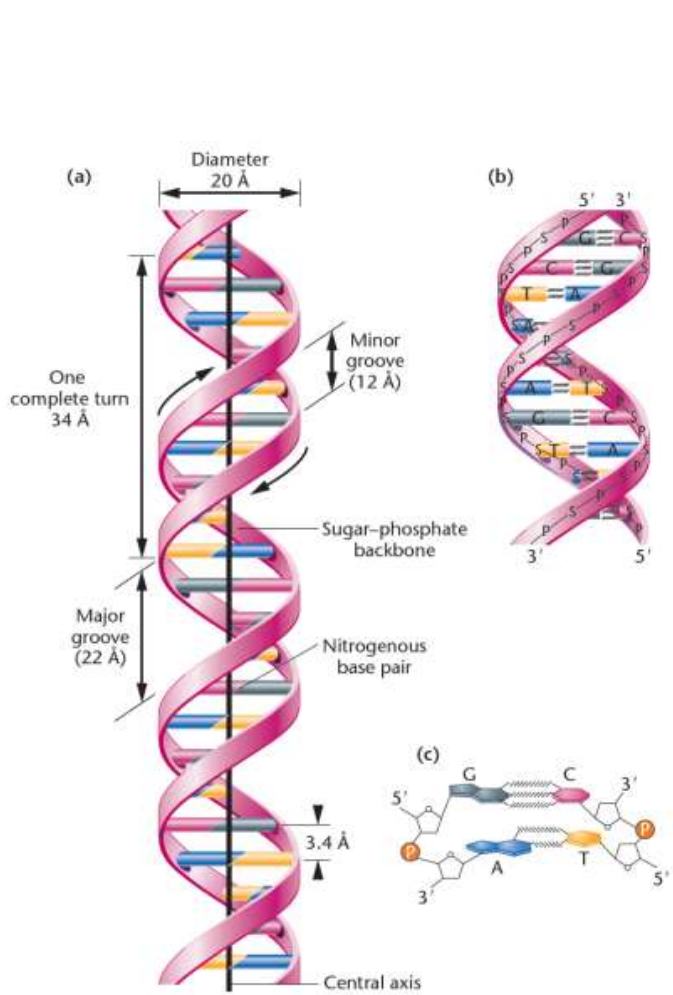
1 complete turn = 360°

10.5 bp per turn

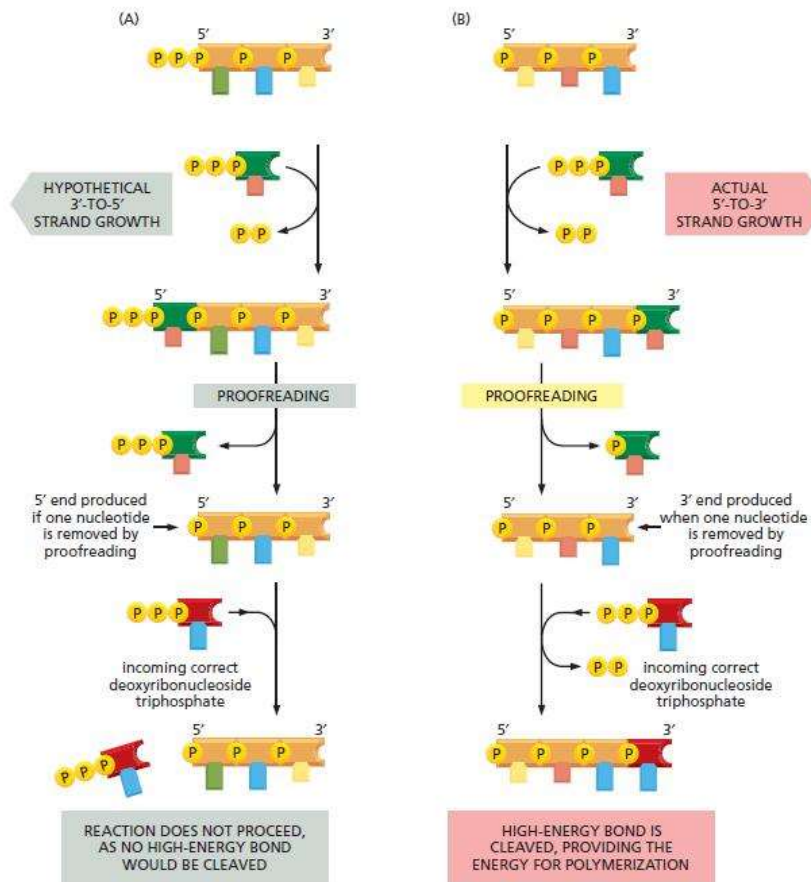
Helix pitch 33.2Å ~34Å

Nucleotide Pairing





Why DNA replication occurs in 5' to 3' direction?



Evolution

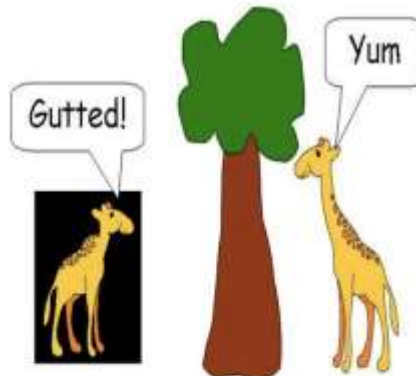
Evolution takes place over millions of years, gradually resulting in new species

Natural Selection

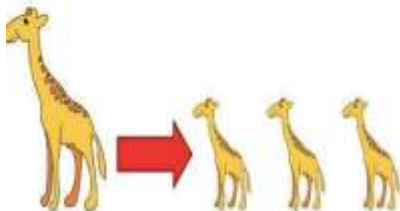
1) Each species shows variation:



2) There is competition within each species for food, living space, water, mates etc



3) The "better adapted" members of these species are more likely to survive - "Survival of the Fittest"

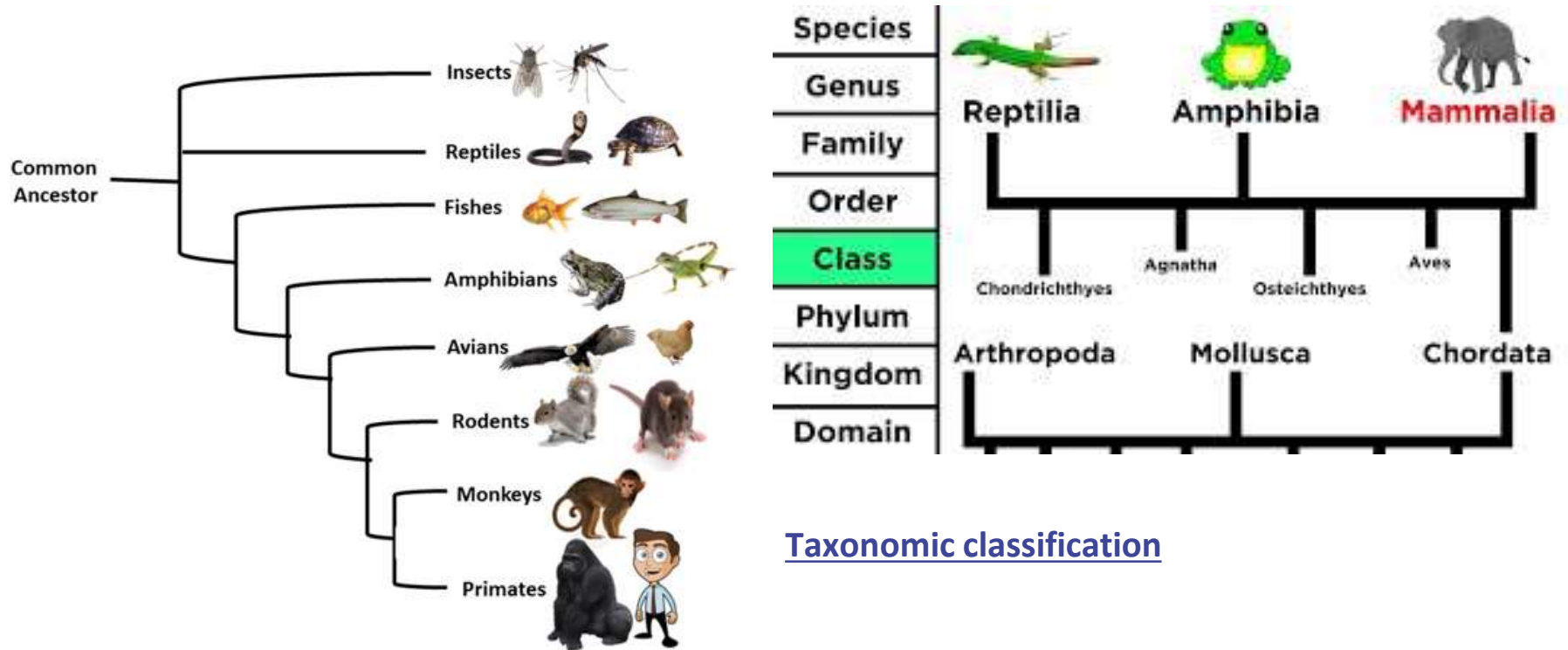


4) These survivors will pass on their better genes to their offspring who will also show this beneficial variation.

It is not the strongest that survive, nor the most intelligent, but the ones most responsive to change.
- Charles Darwin

Evolution: Phylogeny

Phylogeny is the representation of the evolutionary history and relationship between groups of organisms.



Phylogeny helps us to understand the importance of each species in the world.

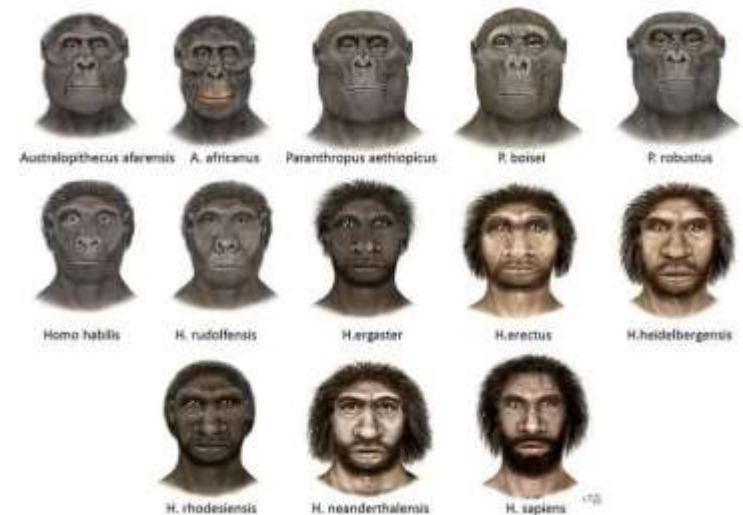
Unity in diversity among life forms

1.2 Million species of plants and animals have been identified

Scientists estimate that the total number of species in the world is 8.7 million

Evolution

Is the reason we have biodiversity in the world.



Extinction

Extinct species are the ones which were considered “NOT FIT” by nature

Although nowadays Human intervention is the main cause of extinction.

Convention on biological diversity stated that almost 150 species are going extinct daily.



Is it possible to bring back an extinct species?

Extinct Bird Re-Evolved Itself Back Into Existence

Once the bird became extinct, it took only 20,000 years for it to return and evolve into a flightless bird again.

By Alexa Leland May 10, 2019, at 3:37 p.m.



RECOMMENDED

These Are the 7 Cheapest Grocery Stores in the U.S.



Don't Use After Financing Deals in August 2022



What Will the Social Security COLA Raise Be for 2024?



China's Degraded Endemic Parrots Near



The Aldabra rail, is the last surviving native flightless bird in the Indian Ocean.